EX:NO:7 Flow control at Data link layer.

Aim:

Write a program to implement flow control at data link layer using sliding window protocol.

Create a sender program:

i) Input window size from the user.

ii) Input a text message from the user.

iii) Consider 1 character per frame.

iv) Create a frame with following fields.

v) Send the frames.

vi) Wait for the acknowledgement from the receiver.

vii) Reader a file called receiver-buffer.

viii) Check ACK field for the acknowledgement number.

Create a receive file:

i) Reader a file called sender-buffer.

ii) check the frame no.

iii) If the frame no are as expected, write the approxiate ACK no in the receiver-buffer file.

Scanned with OKEN Scanner

code;

Sender program;

```
#include <stdio.h>
int main() {
  int window, i, start = 0, ack;
  char msg[100];
  file* fp;
  printf("Enter window size: ");
  scanf("%d", &window);
  int n = strlen(msg);
  while (start < n) {
    fp = fopen("Sender-buffer.txt", "w");
    printf("Sending frames ":);
    for(int i = start; i < start + window && i < n; i++) {
      printf("Frame No: %d, Data: %c\n", i+1, msg[i]);
    }
    fclose(fp);
    fp = fopen("Receiver-buffer.txt", "r");
    if (fp == NULL) {
    printf("Receiver-buffer.txt not found\n");
    continue;
    }
    fscanf(fp, "%d", &ack);
    fclose(fp);
    if (ack == start + window || ack >= n) {
      printf("ACK %d received", ack);
      start += window;
    }
    else {
      printf("Resending frames from frame %d:\n",
                                    start+1);
      }
    }
    return 0;
}
```

Scanned with OKEN Scanner

Receiver program:

```
#include <stdio.h>
#include <unistd.h>
int main() {
    FILE *fp, *ackfp;
    char line[100];
    int frame, expected = 1;
    char data;
    while (1) {
     fp = fopen ("sender_buffer", "r");
     if (fp==NULL)
    {
     printf ("Waiting for sender");
      sleep (2);
      continue;
    }
    while (fgets (line, sizeof(line), fp)) {
    ackfp = fopen ("Receiver buffer", "w");
     if (frame == expected) {
       printf ("Frame %d accepted\n", frame);
        expected ++;
     }
    else {
        printf ("Frame %d rejected", frame);
        }
        fclose (ackfp);
      }
     fclose (fp);
     sleep (3);
  }
   return 0;
  }
```